

Case Study

A 58-year-old man presents to his primary care physician for a routine follow-up visit. He has a 15-year history of type 2 diabetes mellitus and hypertension. His medications include metformin and amlodipine, although he admits to inconsistent adherence. He reports increasing fatigue and swelling in his ankles over the past several months. Physical examination reveals blood pressure of 158/96 mm Hg and bilateral pitting edema of the lower extremities.

Laboratory studies demonstrate:

Tests	Results	Normal Values
Fasting serum glucose	212 mg/dL	70-99 mg/dL
HbA1c	9.4%.	<5.7% mg/dL
Serum creatinine	1.8 mg/dL	0.7-1.3 mg/dL
BUN	32 mg/dL	8-24 mg/dL
Urine albumin	Positive 78 mg/g	<30 mg/g
Urinalysis	Proteinuria without hematuria	pale yellow, clear

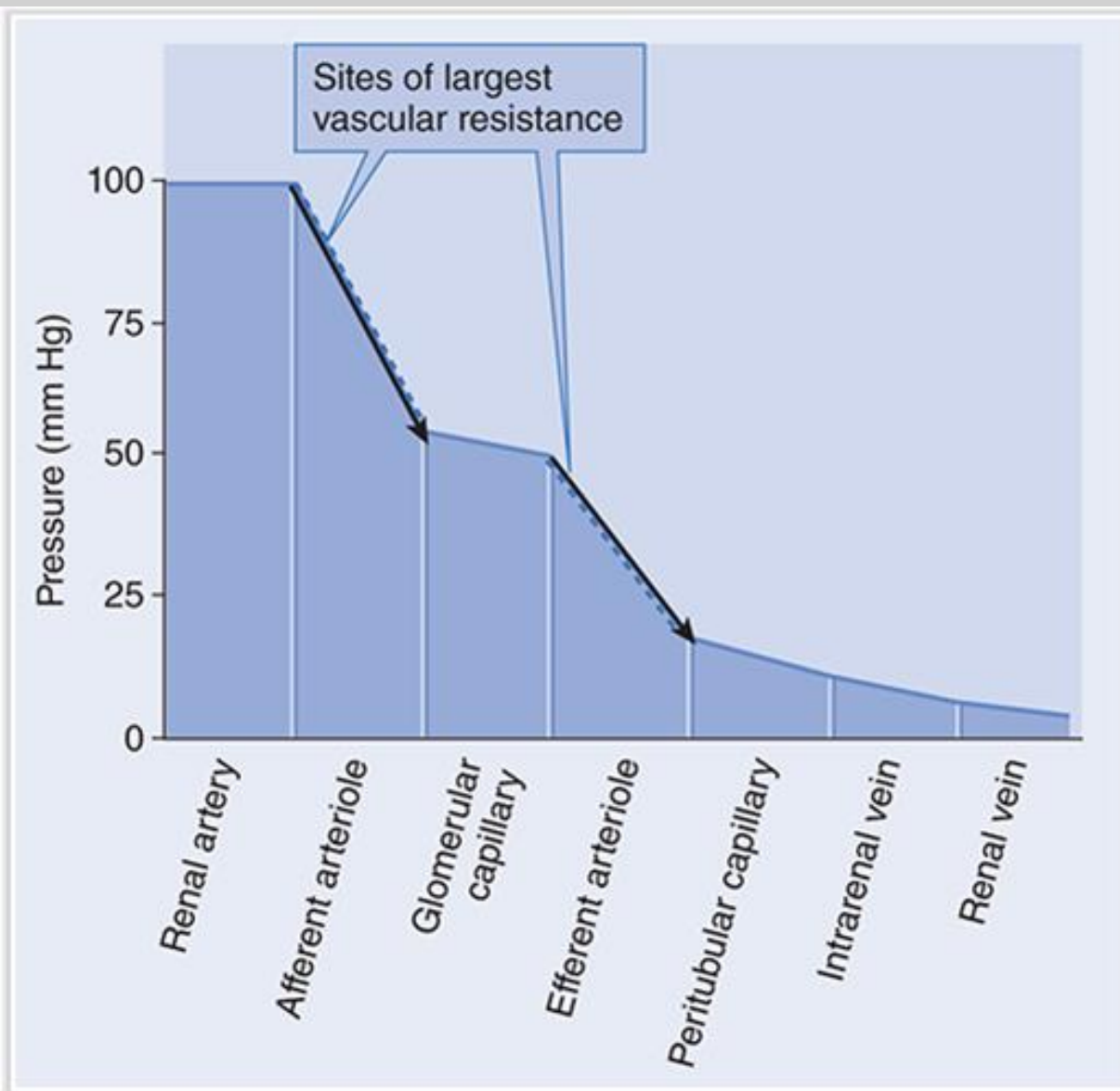
- What is your differential? Why?
- Throughout the disease course, what physiological changes occurred within this patient's renal system? Would a renal biopsy show anything abnormal?
- How are GFR, oncotic & hydrostatic pressures, filtration ability, and autoregulation of blood flow implicated?
- What could have prevented this from occurring? How can you help this patient?

Learning Objectives

1. Demonstrate understanding of the anatomy of the renal system, especially in relation to blood supply, filtration, the renal cortex, medulla, tubular system, and glomerulus.
2. Correlate gross anatomy and function of the kidneys with GFR, resistance, blood pressure, hydrostatic and oncotic pressures.
3. Determine how autoregulation and vasoactive influences affect the kidneys' role in homeostasis.
4. Formulate enough knowledge in filtration and blood flow of the renal system to apply to a pathological process case study.
5. Apply what you have learned to board-style multiple-choice questions.

Resistance Change as Blood Travels into smaller Renal Vessels

- **Net Filtration** in Afferent Arterioles and Glomerular capillary's, **Net reabsorption** in peritubular capillary's
- As blood flows through more resistance (smaller "r"/radius) pressure will **decrease**
- The location of the glomerular capillaries, between the sites of high resistance, results in their having a much higher pressure than the peritubular capillaries.
- **Selective constriction or relaxation of afferent and efferent arterioles allows for sensitive control of hydrostatic pressure (Blood pressure in mm Hg) in intervening glomerular capillary and thus glomerular filtration**



Source: Douglas C. Eaton, John P. Pooler: *Vander's Renal Physiology*, 10e
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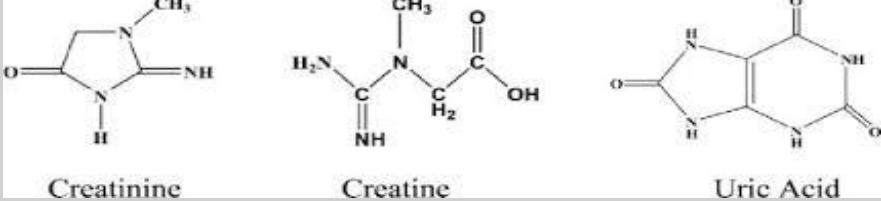
Feature	Renal Blood Flow	Renal Plasma Flow
Definition	Total blood volume passing through kidneys each minute	Volume of cell free plasma passing through the kidneys each minute
Normal value	About 1L/min (About 20% of CO)	600-650 ml/min
Function	Overall oxygen and nutrient delivery to the kidneys	Actual filterable fluid the kidneys will get per minute
Formula	RBF = RPF/(1-Hematocrit)	RPF = RBF*(1-Hematocrit) (HCT) eRPF = $V * U_{pah} / P_{pah}$ = Clearance of PAH

- **Q vs total RBF (difference):**
- **Q:** flow within individual renal vessels (local hemodynamics)
- **RBF:** total blood volume delivered to kidney (global flow)
- **Hematocrit (Hct):**
- % of blood = RBCs
- Normal:
 - Men: 41–50%
 - Women: 36–47%
- $\uparrow$ Hct $\rightarrow$ polycythemia/dehydration
- $\downarrow$ Hct $\rightarrow$ anemia
- $1 - \text{Hct} \approx$ plasma fraction
- $\uparrow$ Hct $\rightarrow \uparrow$ viscosity (η) $\rightarrow \uparrow$ resistance $\rightarrow \downarrow$ renal blood flow
- **PAH (para-aminohippuric acid):**
- Used to estimate **effective renal plasma flow (eRPF)**
- Almost completely cleared from plasma in one pass through kidney
- Clearance of PAH $\approx$ RPF
- Alternatives: radioactive tracers (renal scintigraphy)
- **PAH clearance equation variables:**
- U_{pah} : urine PAH concentration
- V : urine flow rate (mL/min)
- P_{pah} : plasma PAH concentration

More Filtration Formulas to Know

- **Filtration Fraction (FF)** = GFR/RPF = Clearance of creatinine/clearance of PAH
- **Filtered Load** (mg/min) = $GFR \times \text{plasma concentration}$, substance filtered in a given time
- GFR is best estimated with **creatinine** clearance in **urine** ($V \times U/P$)
- eGFR is best estimated with **blood** creatinine
- RPF is best estimated with **PAH** clearance ($V \times U/P$)
- Normal FF is approximately 20%, increasing GFR only increases FF, increasing RPF only decreases FF

The relationship between eGFR and Clearance



- **Creatinine** is the closest endogenous marker to **inulin** for estimating GFR, though it slightly overestimates GFR ($\approx 10\text{--}20\%$) due to tubular secretion.
- Normal serum creatinine is $\sim 0.74\text{--}1.35$ mg/dL in men and $\sim 0.59\text{--}1.04$ mg/dL in women, but varies with muscle mass, age, diet, and activity. It is a byproduct of muscle metabolism (from **creatine**, an energy storage molecule that can also be influenced by supplementation).
- While **inulin** is considered to be the gold standard (freely filtered, not reabsorbed or secreted), creatinine is more practical for routine use.
- **Cystatin C** is an alternative endogenous marker produced by all nucleated cells and is less influenced by muscle mass or diet, but can be affected by conditions such as thyroid disease, diabetes, and smoking.
- Creatinine and cystatin C are the two most used biomarkers for estimating kidney function (eGFR).

Relationship substance "X"	Rationale	Outcome
$C_x < eGFR$	eGFR is higher than clearance (Clearance = $V \cdot U/P$)	Net reabsorption of X or not freely filtered
$C_x > eGFR$	Clearance is higher than eGFR, similar to creatinine (because of minor kidney tubular secretion)	Net secretion of X
$C_x = eGFR$	Clearance and eGFR are equal, similar to Inulin	Not secreted or reabsorbed

Autoregulation

- The Kidneys want to support constant GFR and RBF despite changes in renal perfusion pressures
- Decreased perfusion due to many mechanism (HFrEF, dehydration, renal artery stenosis, glomerular disease etc...) can lead to damage to the kidneys (AKI), especially within the medulla
- Too much perfusion can overwork the kidneys (such as in diabetes, high salt diets, obesity etc...) leading to eventual collapse of renal function
- Mechanisms including **innate smooth muscle tone, Macula Densa cells(chemoreceptors)/JG cells(mechanoreceptors) and the autonomic nervous system regulate** these changes

1. Myogenic Mechanism

- $\uparrow \text{BP} \rightarrow \text{afferent arteriole stretch} \rightarrow \text{vasoconstriction} \rightarrow \downarrow \text{RBF} + \downarrow \text{GFR}$ (proportional; FF unchanged)
- $\downarrow \text{BP} \rightarrow \downarrow \text{stretch} \rightarrow \text{vasodilation} \rightarrow \uparrow \text{RBF} + \uparrow \text{GFR}$

2. Tubuloglomerular Feedback Mechanism

- $\uparrow \text{NaCl at macula densa} \rightarrow \uparrow \text{GFR signal} \rightarrow \text{ATP} + \text{thromboxane A2} \rightarrow \text{afferent constriction} \rightarrow \downarrow \text{GFR} + \downarrow \text{RBF}$
- $\downarrow \text{NaCl at macula densa} \rightarrow \downarrow \text{GFR signal} \rightarrow \text{prostaglandins} + \text{renin (RAAS)} \rightarrow \text{afferent dilation} \rightarrow \uparrow \text{GFR} + \uparrow \text{RBF}$

Vasoactive Substance Influence

- **Vasoconstrictors**

- **Renin**- RAAS system initiating enzyme, made by JG cells in the kidney, catalyst for angiotensinogen (made in liver) to angiotensin 1
- **Angiotensin 2 (Ang 2)**- RAAS system end-product hormone, converted from angiotensin 1 by ACE (made in lung), affects across the body, preferentially constricts the efferent arteriole in the kidneys (greater concentration of AT1 receptors)
- **Norepinephrine (NE)**- Produced in the Locus coeruleus (brain), sympathetic ganglia, and binds to Alpha-1 receptors (G_q)
- **Endothelin (Peptide)**- Produced by kidney endothelial & tubular cells, paracrine/autocrine, imbalance occurs in various pathologies
- **Anti-Diuretic Hormone/Vasopressin (ADH)**- Produced by hypothalamic neurons, stored & secreted by the posterior pituitary (neurohypophysis), acts on the collecting duct V2 receptor, leading to an increase in aquaporins (AQP-2) and increased water reabsorption
- **ATP**- Energy molecule that extracellularly acts on purinergic receptors in the kidney, released by Macula densa cells
- **Thromboxane A₂ (TxA₂)**- Lipid mediator derived from COX enzymes/arachidonic acid that is involved in platelet aggregation, increased release in inflammatory conditions, made by platelets, macula densa cells, podocytes, and endothelial cells, G_q mechanism
- **High dose Dopamine**- D1 and D2 receptor activation as well as α_1 adrenergic receptors (G_q ; Ca^{++} influx)
- **Cocaine and Methamphetamine**- Block NE and dopamine re-uptake, increasing α -1 (G_q ; stimulation, increase endothelin and decrease nitric oxide

• Vasodilators

- **Prostaglandins (PGE₂/PGI₂)-** Synthesized locally by kidney cells COX-1 and COX-2 enzymes, increase cAMP, during stress and volume depletion save kidney from ischemia (act as a "shield"), relax mesangial cells and afferent arteriole preferentially
- **Nitric Oxide-** Keeps a basal level of relaxation in kidney vessels, Comes from amino acid L-arginine, diffusible gas across cells, decreases calcium release, and increases smooth muscle relaxation, without it there would be immense renal vasoconstriction
- **Atrial Natriuretic Peptide (ANP/BNP)-** Made by atrial myocytes within the heart, decreases total body salt and water
- **Kallikrein/Bradykinin-** Produced in the distal nephron, leads to Prostaglandin and NO local release
- **Low dose Dopamine-** Acts on D1 and D2 receptors (cAMP pathway leading to PKA and vasodilation) **without** interacting with Alpha-1 receptors

Think about each of these substances and how increasing them or decreasing them would affect kidney physiology

For example, what effects would something that inhibits ACE (Lisinopril) have on GFR, Blood pressure and the RAAS system?

How would something that decreases prostaglandins (NSAIDs) effect tone of afferent and efferent arterioles?

What would blocking V2 receptors do?

Pathologies related to Filtration and Blood Flow

- Glomerulonephritis/Nephritic Syndrome
- Heart Failure
- Liver Cirrhosis
- Multiple Myeloma
- Amyloidosis
- Chronic Kidney Disease
- Dehydration and Hemorrhage
- Diabetic Nephropathy
- Nephrotic Syndromes
- Infectious disease- Pyelonephritis, TB, Malaria, Leptospirosis etc...
- Chronic Hypertension
- Acute Kidney Injury
- Goodpasture
- Rhabdomyolysis
- Nephrotoxic Drugs- NSAIDS, Lithium, Amphotericin B, Acyclovir, Cisplatin, Allopurinol etc..

Vander's Textbook Questions

- 2–1. *Blood enters the renal medulla immediately after passing through which vessels?*
 - *Arcuate arteries.*
 - *Peritubular capillaries.*
 - *Afferent arterioles.*
 - *Efferent arterioles.*
- 2–2. *Which cell type is the main determinant of the filterability of plasma solutes?*
 - *Mesangial cell.*
 - *Podocyte.*
 - *Endothelial cell.*
 - *Vascular smooth muscle.*
- 2–3. *Which one of the following is NOT subject to physiological control on a moment-to-moment basis?*
 - *Hydrostatic pressure in glomerular capillaries.*
 - *Selectivity of the filtration barrier.*
 - *Renal blood flow.*
 - *Resistance of efferent arterioles.*
- 2–4. *A substance is freely filtered and has a certain concentration in peripheral plasma. You would expect the substance to have virtually the same concentration in*
 - *the glomerular filtrate.*
 - *the afferent arteriole.*
 - *the efferent arteriole.*
 - *All of these places.*
- 2–5. *In the face of a 20% decrease in arterial pressure, GFR decreases by only 2%. What could account for this finding?*
 - *The resistances of the afferent and efferent arterioles both decrease equally.*
 - *Glomerular mesangial cells contract.*
 - *Efferent arteriolar resistance increases.*
 - *Afferent arteriolar resistance increases.*

Board Style Questions

1. In a novel drug clinical trial, a healthy 24-year-old male volunteer receives an infusion of a drug in the NSAID class that has not been FDA approved for release. This drug has been previously proven to have a similar MOA to ibuprofen. The purpose of this trial is to see how the drug can affect renal physiology. The participant is 5'10, 195 pounds, current HR is 75 bpm, and blood pressure is 132/85 mmHg. All other vitals are within normal limits (WNL). He takes no medications and reports NKDA. He is in good health with no medical or psychiatric conditions and admits to having no past surgical history except an uncomplicated adenoidectomy at 12-years old. He currently attends graduate school, sleeps 6-7 hours a night and has no family history of kidney disease. Based on knowledge of how typical NSAIDs effect kidney physiology, which of the following changes is most likely to occur in this participant's kidneys after administration of this new medication?

Renal Blood Flow

Glomerular Capillary Hydrostatic Pressure

GFR

- | | |
|----|-----------|
| A. | ↑ |
| B. | ↑ |
| C. | ↓ |
| D. | ↓ |
| E. | No change |

↑

↓

↓

↑

No change

2. A healthy 28-year-old man with a previous history of GERD and GAD is found to have a plasma oncotic pressure of 32 mmHg, a glomerular capillary hydrostatic pressure of 60 mmHg, a Bowman's space hydrostatic pressure of 18 mmHg, and a Bowman's space oncotic pressure of 0 mmHg. What is the net filtration pressure driving GFR?

- A. 10 mmHg
- B. 14 mmHg
- C. 42 mmHg
- D. 28 mmHg
- E. 60 mmHg

3. A 34-year-old female presents to the ED with recurrent panic attacks. She states that early that day, she had one attack so intense she couldn't leave her house to get to work. Despite lasting 10 minutes, it was so exhausting and debilitating that she felt she needed to get emergency care before trying to return to work the following day. At the time of arrival, her vitals are within normal limits (WTN); she is alert and oriented X 3 and shows no signs of distress but some fatigue. Suddenly, while being interviewed by a resident physician, she starts to sweat, her pupils dilate, and stops answering the residents' questions. She starts to visibly show discomfort and appears in acute distress. Her mean arterial pressure suddenly rises from 90 mm Hg to 130 mm Hg. Despite the increase in MAP, her kidneys keep her GFR relatively stable. Which homeostatic mechanism is primarily responsible for keeping her GFR constant?

- A. Increased aldosterone secretion
- B. Activation of the renin-angiotensin system
- C. Tubuloglomerular feedback and myogenic response
- D. Increased sympathetic stimulation
- E. Increased ADH release

4. While studying snake renal physiology, you discover a new vasoactive agent which does not exist in mammals. This molecule is a peptide hormone, and you decide to name it "Slither 1". It is figured out that "Slither 1" among other things, preferentially constricts the efferent arteriole. Which hormone acts an analogous way in humans and what hemodynamic changes would be expected at low-to-moderate "Slither 1" levels in the snake's kidney (assuming snake physiology is identical to human)?

- A. Angiotensin II, decreased glomerular capillary hydrostatic pressure and decreased GFR
- B. Angiotensin II, increased glomerular capillary hydrostatic pressure and increased GFR
- C. Angiotensin I, increased renal blood flow with no change in GFR
- D. ANP, decreased filtration fraction with increased plasma flow
- E. Angiotensin II, decreased oncotic pressure in glomerular capillaries

5. A 32-year-old female ice hockey goalie was playing in WHL finals when an opposing player accidentally tripped, colliding into her and slicing her jugular vein with her skate blade. An EMT and paramedic bystander in the crowd rushed to her aid but could only compress hard to enough to get her in the ambulance and rush her to the nearby hospital. At the hospital, her pulse is thready, but around 150 bpm, BP is 82/59 mmHG, RR is 38 breaths/min. She is cold and clammy, confused and lethargic. As she's being treated, she is also found to have acute oliguric renal failure, with a non-physiological drop in GFR. Which combination of vasoactive effects best explains the drop in GFR?

- A. Prostaglandin E₂ dilation of afferent arteriole and NO release
- B. Sympathetic-mediated afferent arteriolar constriction and high angiotensin II levels causing efferent constriction that overcomes glomerular pressure
- C. CANP-mediated efferent constriction and afferent dilation
- D. Dopamine-mediated vasoconstriction of both arterioles equally
- E. Endothelin dilation of the efferent arteriole

6. A 24-year-old man presents with hemoptysis, fatigue, and dark-colored urine for 2 weeks. Physical examination is notable for mild hypertension. Laboratory studies show elevated serum creatinine and numerous red blood cell casts on urinalysis. A kidney biopsy proves crescent formation in Bowman's space. Immunofluorescence reveals a linear pattern of IgG deposition along the glomerular basement membrane.

The patient's renal disease is caused by autoantibodies directed against a major part of which of the following structures involved in the glomerular filtration barrier?

- A. Fenestrated endothelial cells of the glomerular capillaries
- B. Type IV collagen within the glomerular basement membrane
- C. Podocyte foot processes forming the slit diaphragm
- D. Mesangial cells supporting the glomerular capillaries
- E. Proximal tubular epithelial cells responsible for protein reabsorption